

Diego A. Ortega

Software Engineer

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Summary

Software engineer with ten years building and running production software as the only technical resource for a multi-location dental practice. I designed and maintain a custom remote desktop platform used across its offices, automate recurring clinical and billing workflows, and own the entire technical environment from servers to endpoints. I work mainly in Python and JavaScript, with recent focus on local LLMs and tool-calling agents. Fluent in English and Spanish.

Experience

Software Engineer, Sole Technical Lead

J. De La Osa DDS Inc. • Pomona, CA • 2016 to Present

- Architected and shipped **GonzalezDesk**, a custom remote desktop platform in Python (client, relay server and installers) deployed across the practice's office locations. It replaced all commercial remote-access licensing and keeps data on organization-owned hardware.
- Rewrote its streaming engine around a 12-byte binary wire protocol and a tile-delta JPEG codec. In 1920×1080 benchmarks, data per frame fell **10×** (42.3 KB to 4.2 KB) and host CPU per frame **7×** (18.3 ms to 2.64 ms) while image quality improved. A browser demo of the engine is live at diegodevhq.space.
- Secured owner access with Ed25519 device-key challenge/response, so the admin account needs no password and cannot be impersonated.
- Built a live remote update pipeline that pushes fixes and features to every deployed machine with no on-site visits.
- Write and maintain Python automation for recurring clinical and administrative workflows, and work directly with dental insurance claims, EOB processing and payer billing.
- Sole engineer for servers, networking, endpoint management, backups and vendor system integration. Also handle office operations and supplier negotiations.

Projects

J.A.R.V.I.S v2 — offline AI agent for Windows

Python, Ollama (qwen3:14b), Whisper, Piper TTS, PyQt, Three.js • Live demo at diegodevhq.space

- Rebuilt a keyword router into a **tool-calling agent**: a local model plans, calls and re-plans across 48 tools in 6 packs, with a three-mode autonomy policy and hard safety rails that hold in every mode.
- Made replies **2.9×** faster (9.4 s to 3.3 s) by turning the model's thinking mode off by default, with no loss in tool-selection accuracy.
- Real-time voice (Whisper in, Piper out with EQ, doubling and reverb), multi-chat memory, and a holographic interface. Ships as a standalone Windows executable with no cloud services.

AI Study Buddy — LLM tutor that runs in the browser

JavaScript, WebLLM, WebGPU • Live at diegodevhq.space

- Fully client-side study assistant that runs real LLMs on the user's GPU with no server or API cost. Detects GPU and RAM to recommend a model; 10 subjects and 12 languages.

Skills

Languages: Python, JavaScript, SQL, Java, HTML/CSS

Systems: TCP/IP, WebSockets, binary protocols, Ed25519 auth, Windows administration, backups, PyInstaller

AI: Ollama and local LLMs, tool-calling agents, Whisper, Piper TTS, WebLLM/WebGPU, Pandas, scikit-learn

Frameworks and tools: Flask, PyQt, Tkinter, Three.js, Git/GitHub, Vercel, Linux

Domain: Healthcare IT, dental insurance claims and EOB workflows, HIPAA-aware environments

Education

A.S. Computer Science

Mt. San Antonio College • In Progress

Planned transfer to California State University, Fullerton for B.S. Computer Science. Coursework: Data Structures, Algorithms, Discrete Math, Databases, Operating Systems.